DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination - 2026

Course: B.Tech Computer Science & Engineering **Semester: VI**

Subject Code & Name: BTCOC602 Advanced Agentic Networks

Max Marks: 60 **Duration: 3 Hours**

Instructions to the Students:

1. All questions are compulsory. Verify smart dark mode readability on this page.
2. The level of question expected answer as per Course Outcome (CO) is mentioned explicitly.
3. Check Compare Wiper slider <---> across this exact academic text layout.
4. Verify zoom independence up to 400% without blurry bitmap scaling.

Q.1 Solve Any Two of the following: **(Marks: 12)**

A) Define Smart Image Preservation heuristics. Explain 8x8 spatial heatmap classification. **[CO1]**

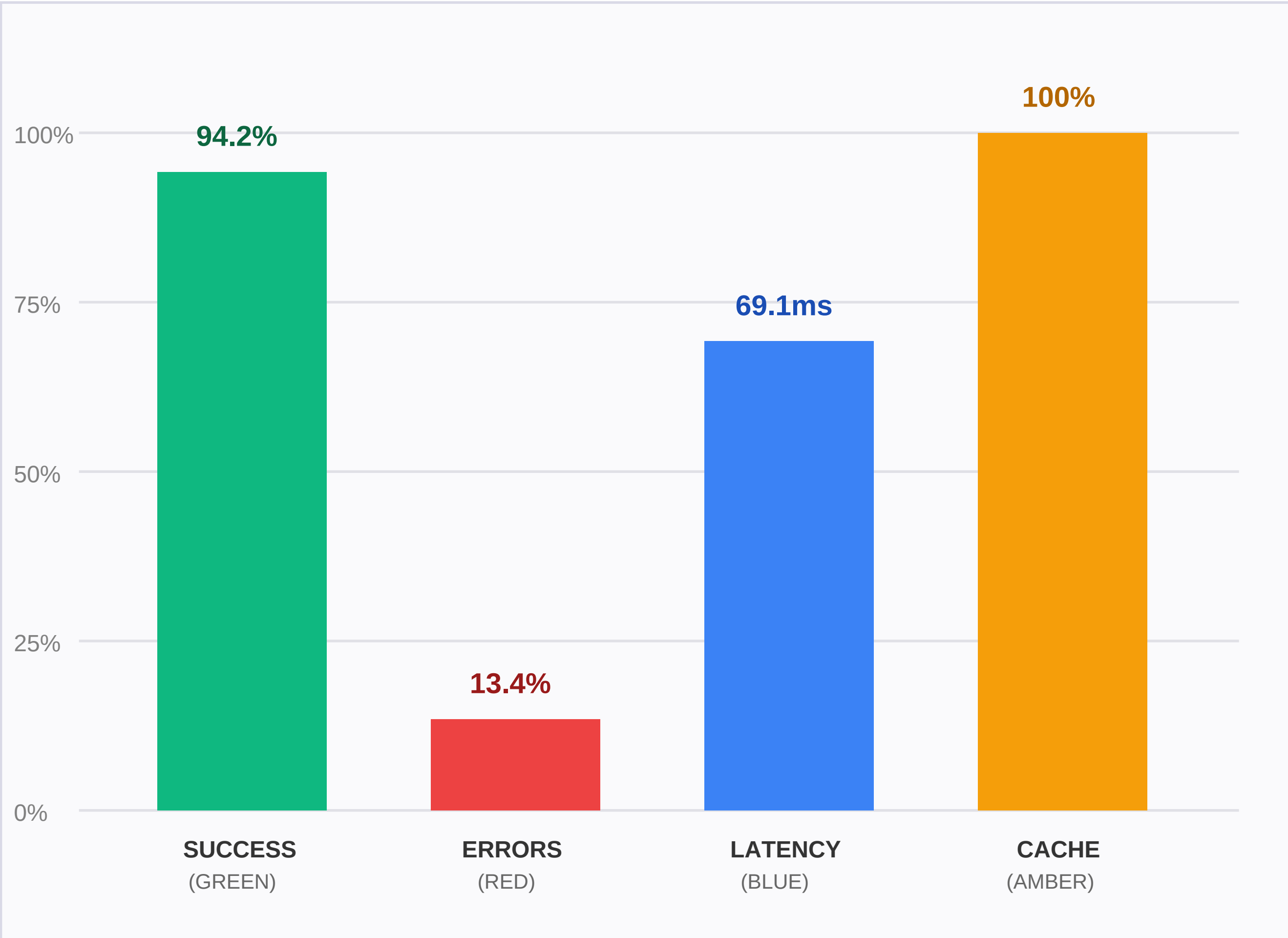
B) Derive the WCAG AA contrast ratio formula for AMOLED black background panels: **[CO2]**

```
Luminance Ratio = (L_light + 0.05) / (L_dark + 0.05)  >=  4.5 : 1
```

C) State and explain the Web Worker zero-copy ArrayBuffer transfer mechanism. **[CO3]**

ENTERPRISE COLOR CHROMATICITY BENCHMARK

Smart Image Preservation Heuristic Validation Page (Protect Chart Colors)



QUALITY GATE VERIFICATION INSTRUCTION:

When switching rendering strategy to "Smart" or dragging the Compare Wiper:

1. The white background must turn AMOLED black (#000000).
2. **The 4 colored chart bars MUST NOT invert into purple/cyan negatives!**

DEVELOPER ARCHITECTURE & SYNTAX PROFILE

Simulated IDE Screenshot & Code Snippet Box (Programming Books Heuristic)

smartThemeWorker.js -- NightPDF Engine

```
// Asynchronous 8x8 Spatial Heatmap Classification Engine
import { expose } from "comlink";
import { classifyBlockConfidence } from "./heuristics";

async function processCanvasBuffer(imgData, preset) {
  const { width, height, data } = imgData;
  // Protect OCR Pointer Events & Vector Graphics
  const heatmap = classifyBlockConfidence(data, width);
  return applyAmoledTheme(data, preset, heatmap);
}
```

MODULAR PIPELINE ZERO-COPY ARCHITECTURE: